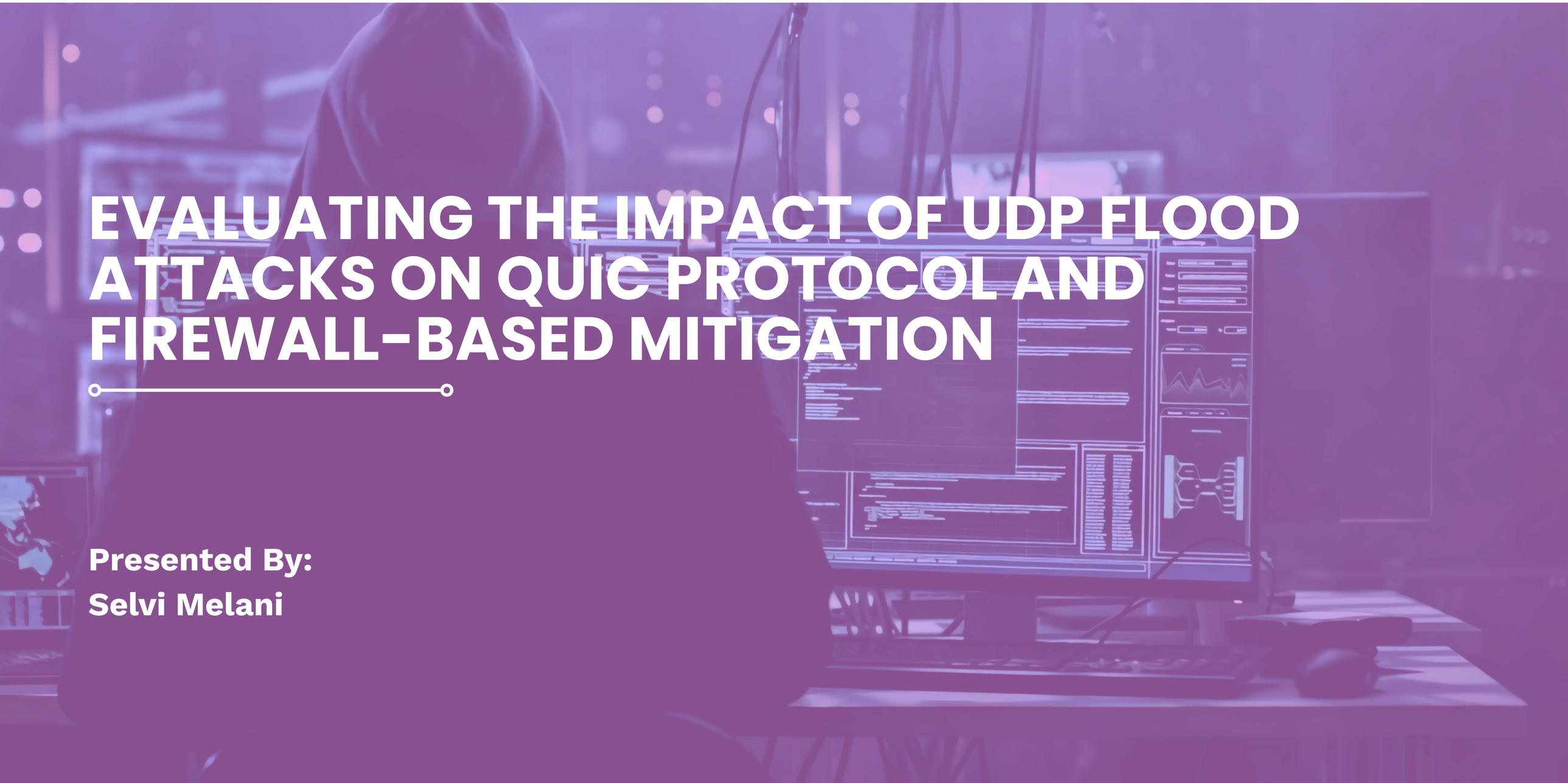


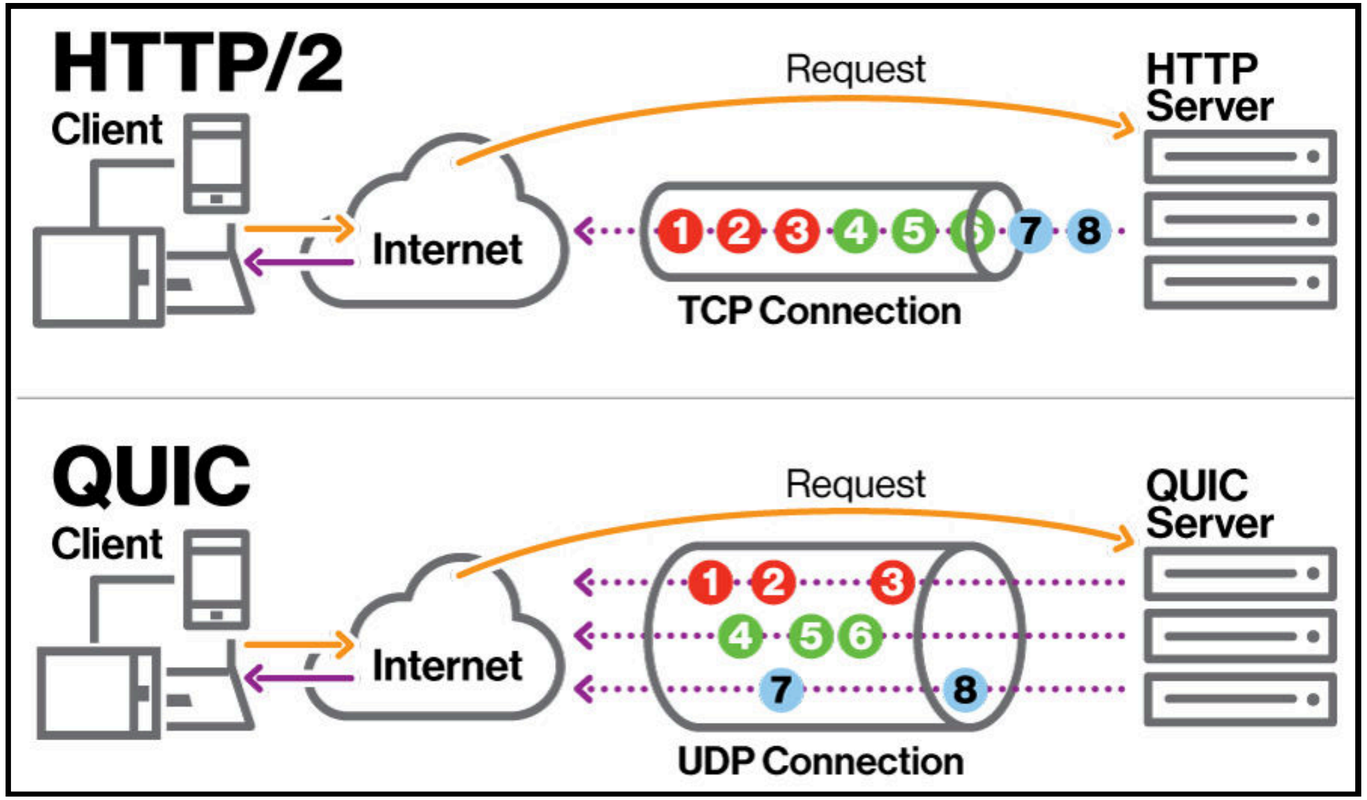
EVALUATING THE IMPACT OF UDP FLOOD ATTACKS ON QUIC PROTOCOL AND FIREWALL-BASED MITIGATION

Presented By:
Selvi Melani

INTRODUCTION

QUIC

- QUIC is **a modern UDP-based transport protocol.**
- **QUIC is faster than TCP** because it eliminates latency during the handshake.
- "*Because QUIC uses UDP as its substrate, endpoints are exposed to the same **denial-of-service** considerations as any UDP-based protocol.*" — RFC 9000, Section 21 (IETF, 2021).
- Therefore, **it is important to test QUIC against UDP**



Resource: Devopedia "QUIC." Version 5, (CC BY-SA 4.0)

OBJECTIVE

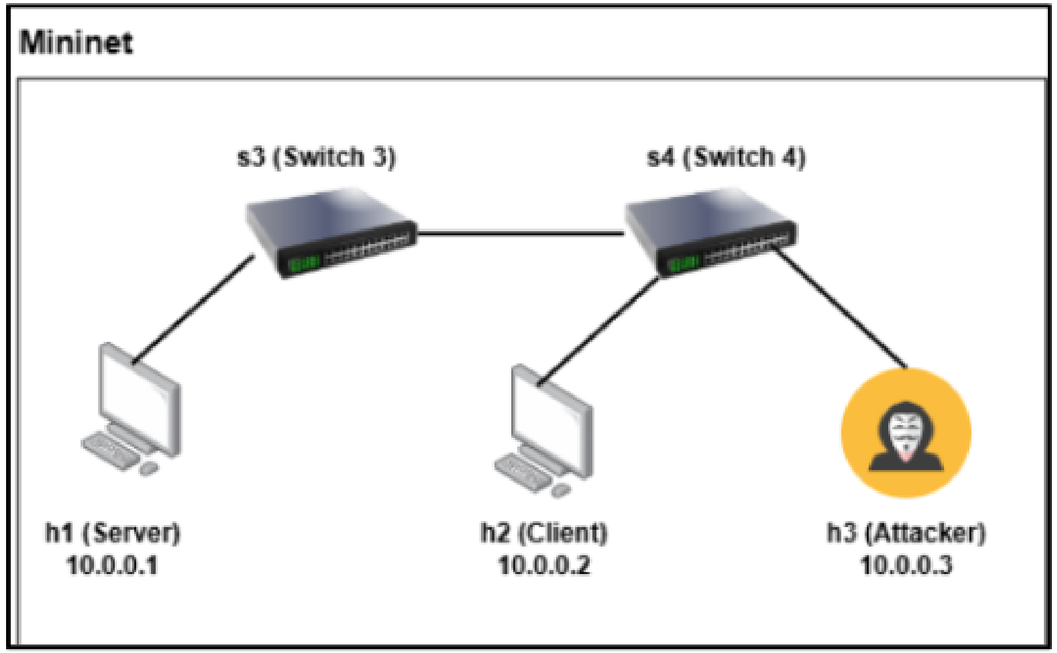
Implementing QUIC in Mininet → Testing UDP-based DoS attacks → Analyzing its impact on the server → Designing a firewall defense system → Comparing the situation before and after the defense

INTRODUCTION

MININET

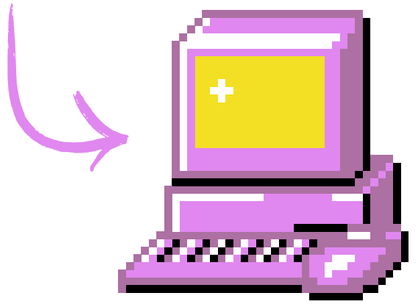
- **Mininet** is an SDN network emulator that c**an simulate the behavior of end hosts, switches, routers, controllers, and links within a Linux kernel** (Mellisa, 2018).

TOPOLOGY



INITIATION

- **Script custom topology in Mininet:**
- Run a custom topology using the command: '**sudo mn --custom topo2.py --topo mytopo'**



```
 GNU nano 7.2
from mininet.topo import Topo

class MyTopo(Topo):
    def build(self):

        # Hosts
        h1 = self.addHost('h1')  # QUIC Server
        h2 = self.addHost('h2')  # QUIC Client
        h3 = self.addHost('h3')  # Attacker

        # Switches
        s3 = self.addSwitch('s3')
        s4 = self.addSwitch('s4')

        # Links
        self.addLink(h1, s3)
        self.addLink(s3, s4)
        self.addLink(s4, h2)
        self.addLink(s4, h3)   # Attacker connected here

topos = {
    'mytopo': (lambda: MyTopo())
}
```

Connectivity Testing:

- Ping tests between hosts indicate that the **connection is functioning normally.**
- This confirms that the topology is ready for the QUIC experiment.

```
mininet> pingall
*** Ping: testing ping reachability
h1 -> h2 h3
h2 -> h1 h3
h3 -> h1 h2
*** Results: 0% dropped (6/6 received)
mininet> nodes
available nodes are:
h1 h2 h3 s3 s4
```

QUIC ENVIRONMENT SETUP

DOWNLOAD AIOQUIC, CREATE, AND ACTIVATE A VIRTUAL ENVIRONMENT

- **Aioquic is available on GitHub: https://github.com/aiortc/aioquic.**
  - To download it, use the command `git clone https://github.com/aiortc/aioquic.git'.
- **The purpose of creating and activating a virtual environment is to:**
  - Ensure that QUIC dependencies do not interfere with the Linux system.
  - All libraries (aioquic) run within this environment.

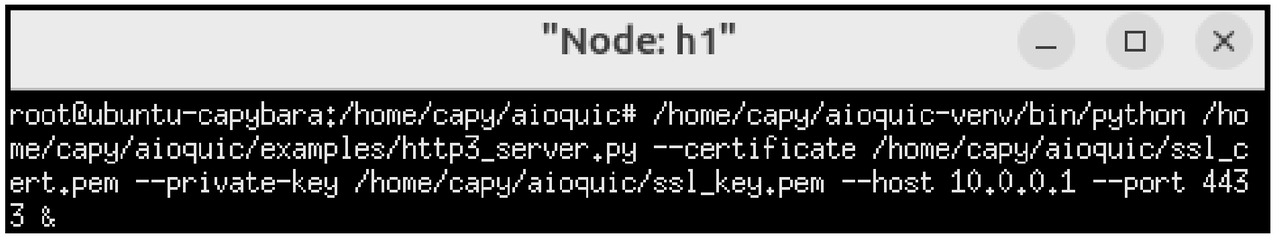
```
capy@ubuntu-capybara:~/aioquic$ source /home/capy/aioquic-venv/bin/activate
```

- Once that was successful, **aioquic was installed on Mininet.**

RUNNING A QUIC SERVER ON H1

- **Running the QUIC Server on h1:**

```
mininet> xterm h1
```



RUNNING THE QUIC CLIENT ON H2

- **Running the QUIC Client on h2:**

```
mininet> xterm h2
```

```
mininet> h2 /home/capy/aioquic-venv/bin/python /home/capy/aioquic/examples/http3_client.py
 --ca-certs /home/capy/aioquic/ssl_cert.pem https://10.0.0.1:4433/
```

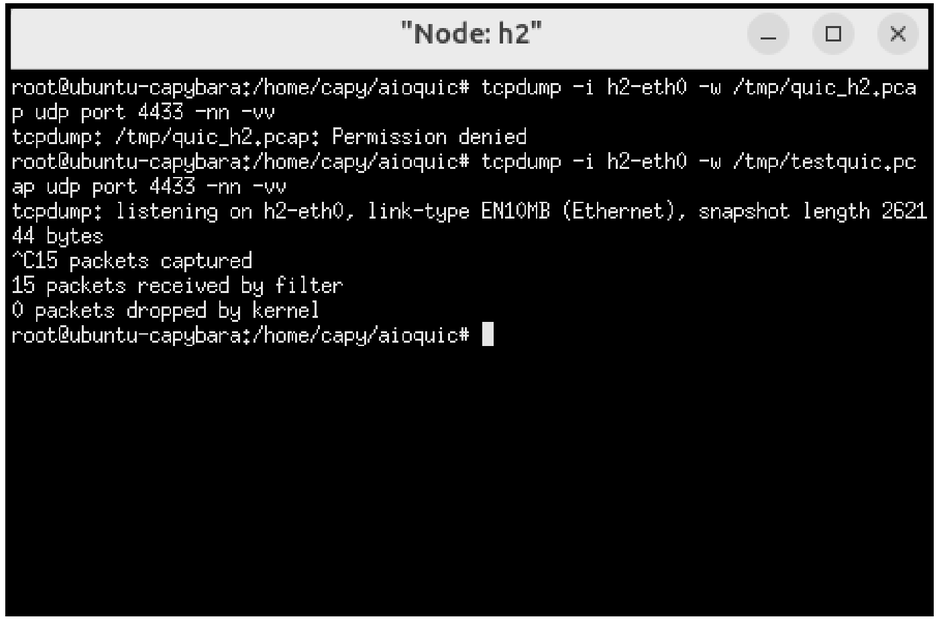
- To run the server, use the http3_server.py command.
- To run the client, use the http3_client.py command.

QUIC CLEAN CONDITION

01

Capture Traffic

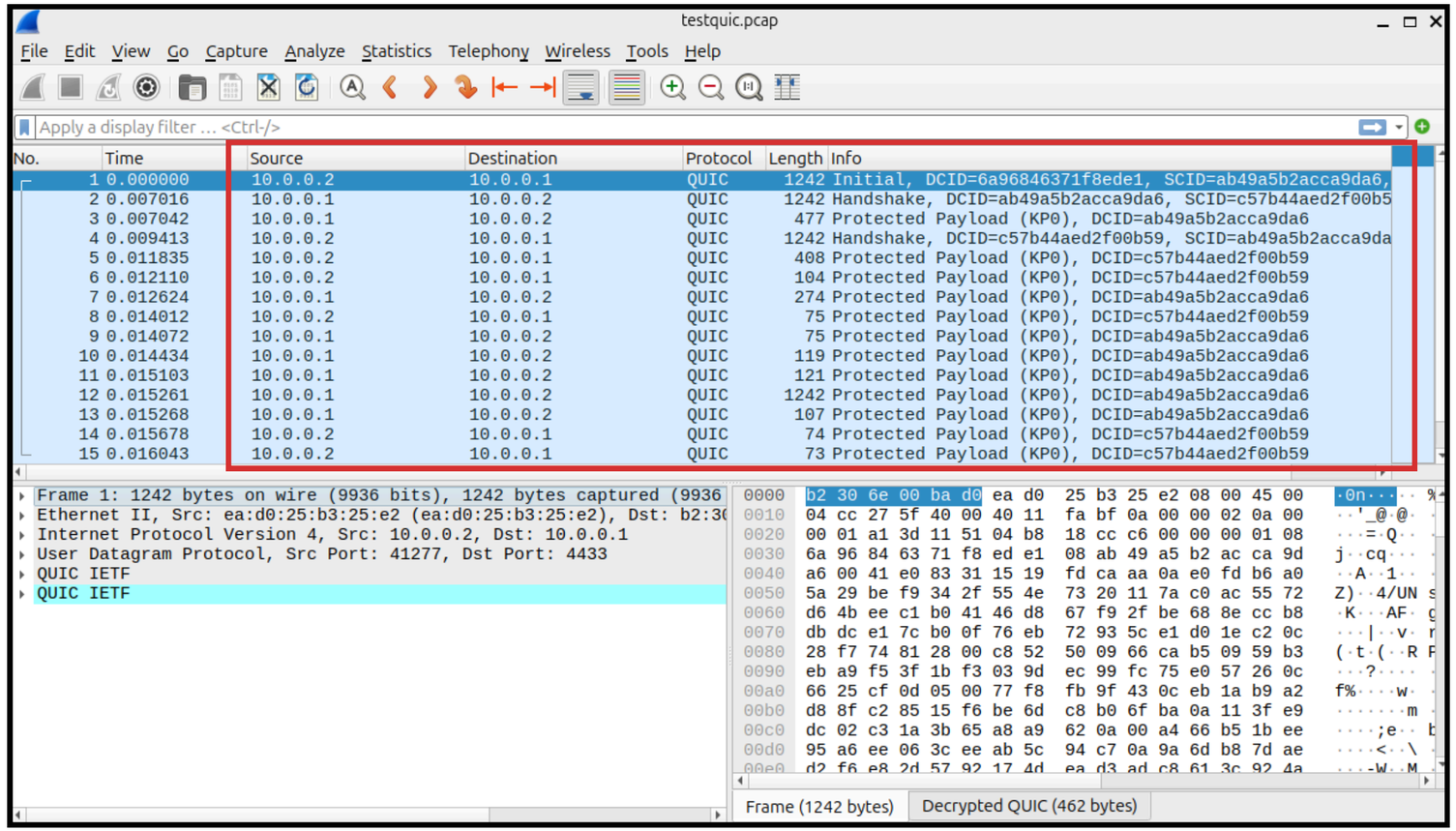
- Capture is running on host h2 (client).
- Saves QUIC packets to the testquic.pcap file for analysis in Wireshark.



- **Objective**
  - Monitoring QUIC traffic under normal conditions.
  - Serving as a baseline for comparing conditions after a UDP flood attack.

02

- **QUIC follows the normal process: Initial → Handshake → Encrypted Payload.**
- There are no abnormal packets, spoofing, or traffic floods.
- All payloads are encrypted (Protected Payload), indicating that QUIC is operating securely.



ATTACK SCENARIO

UDP FLOOD ATTACK

- **UDP Flood Attack → A DoS attack that involves sending a massive number of UDP packets to a specific port to flood the server.**
- QUIC uses UDP → so QUIC servers are highly vulnerable to this attack.
- Test Objective → **To observe the QUIC traffic response when the server receives a flood of UDP packets.**
- In this experiment, the UDP Flood attack was carried out using the 'hping3' command. All attacks were carried out from host h3 (attacker) to the QUIC server on host h1 (10.0.0.1) on UDP port 4433, with a fixed duration of ±10 seconds in each scenario.

Scenario 1: Intensity Variation (Packet Rate)

Scenario	Intensity	Command to Attack	Objective
A1	Low	hping3 –udp -p 4433 –interval u20000 10.0.0.1	Minor disruption
A2	Medium	hping3 –udp -p 4433 –interval u5000 10.0.0.1	Moderate Load
A3	High	Hping3 –udp -p 4433 –flood 10.0.0.1 -p 4433	DoS Behavior

01

Scenario 2: Payload Variation (UDP Length)

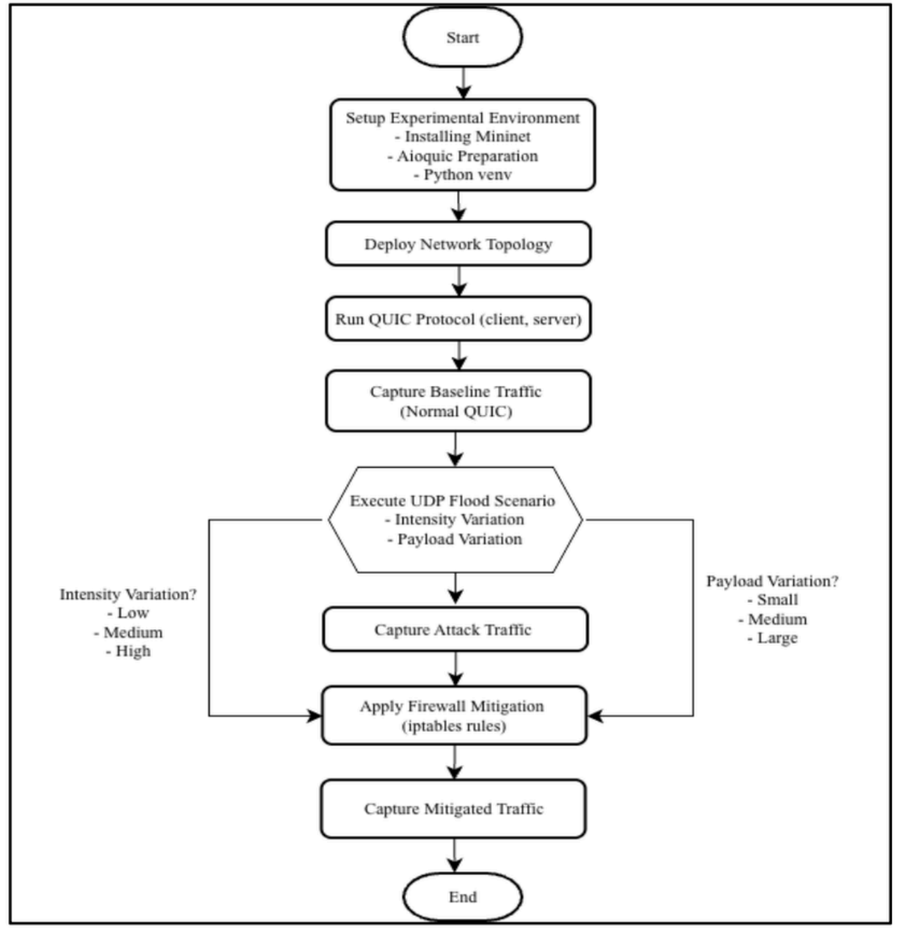
Scenario	Intensity	Command to Attack	Objective
B1	Small	hping3 –udp -p 4433 -d 10 –interval u5000 10.0.0.1	Minor disruption
B2	Medium	hping3 –udp -p 4433 -d 200 –interval u5000 10.0.0.1	Moderate Load
B3	Large	hping3 –udp -p 4433 -d 800 –interval u5000 10.0.0.1	DoS Behavior

02

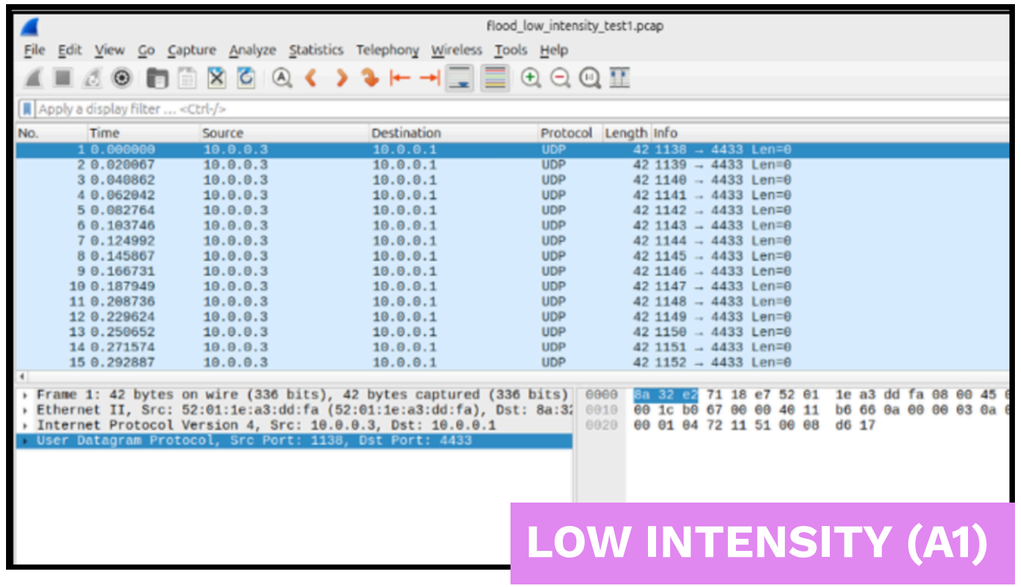


FLOWCHART

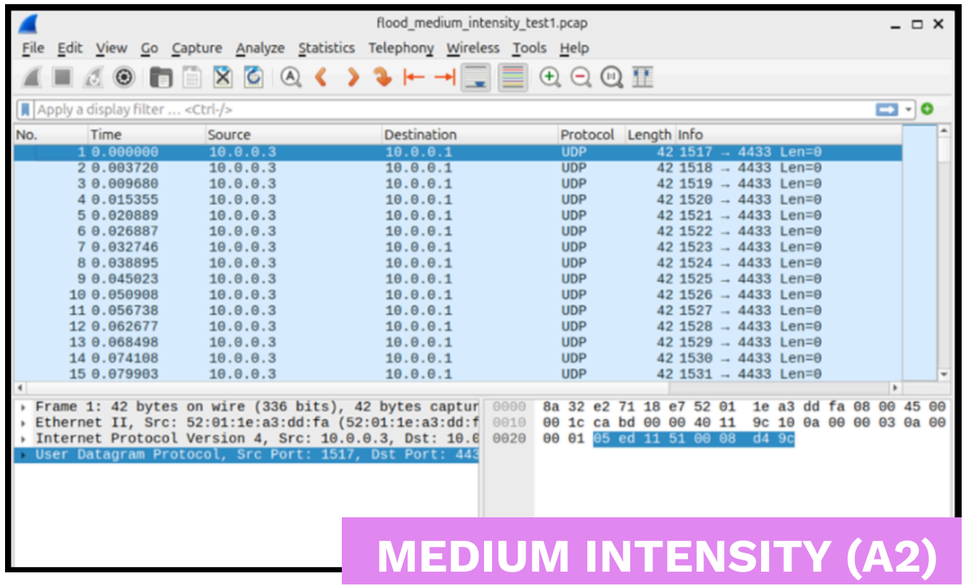
(Summary of Methodology)



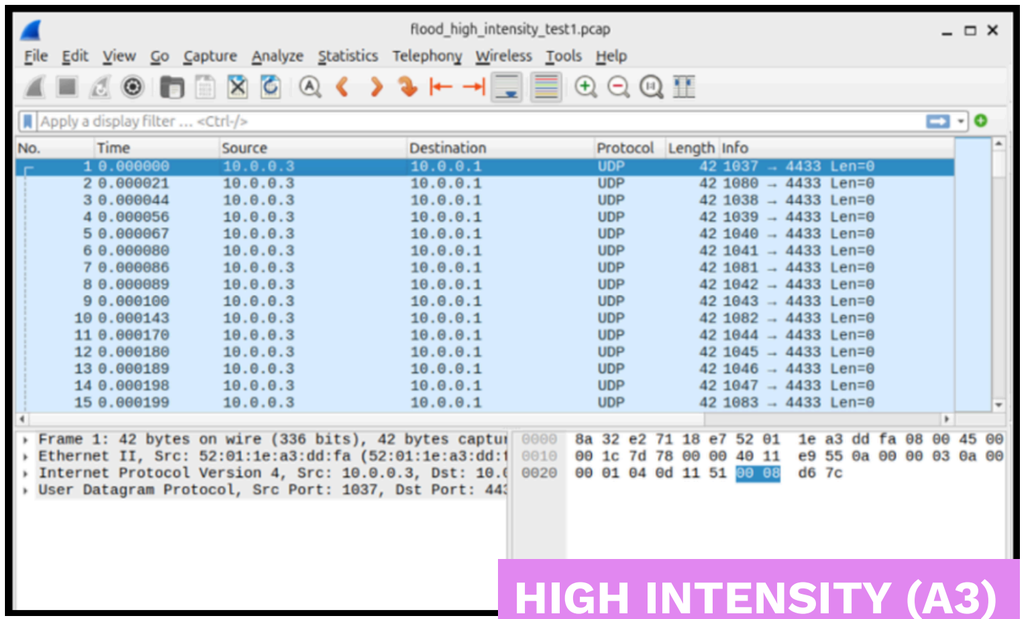
RESULTS OF SCENARIO 1 ATTACK



LOW INTENSITY (A1)

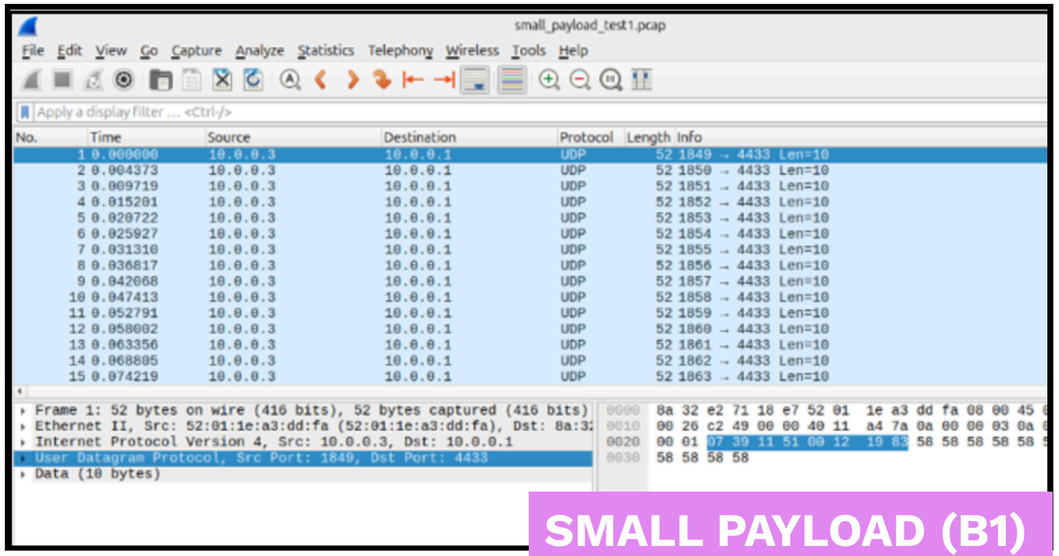


MEDIUM INTENSITY (A2)

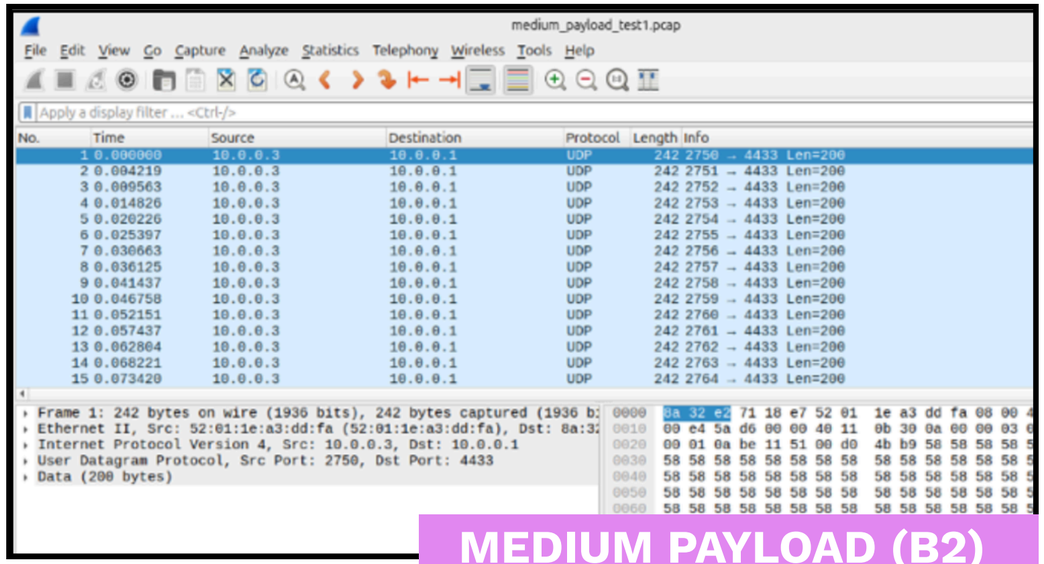


HIGH INTENSITY (A3)

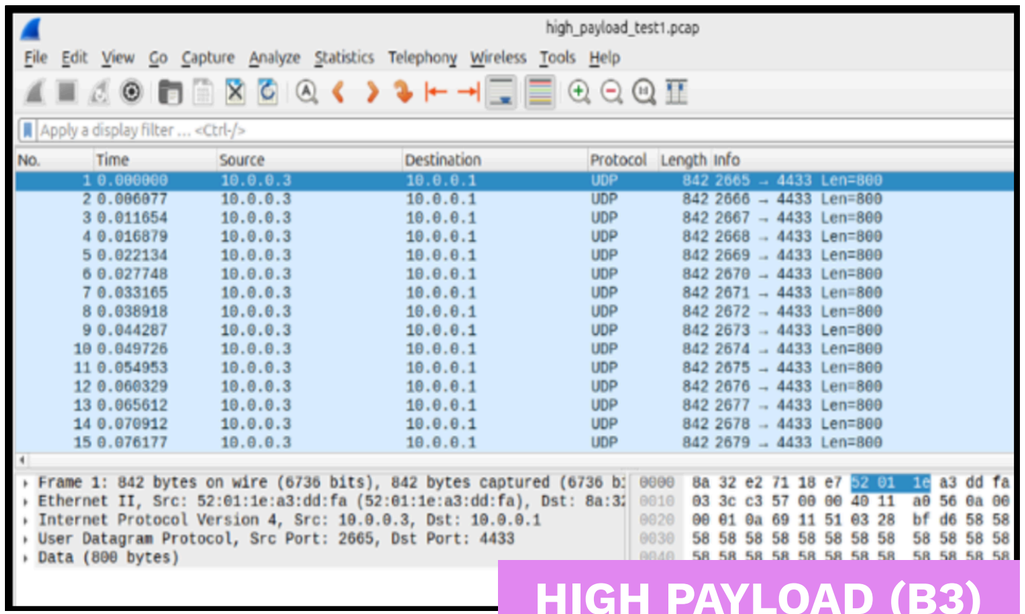
RESULTS OF SCENARIO 2 ATTACK



SMALL PAYLOAD (B1)



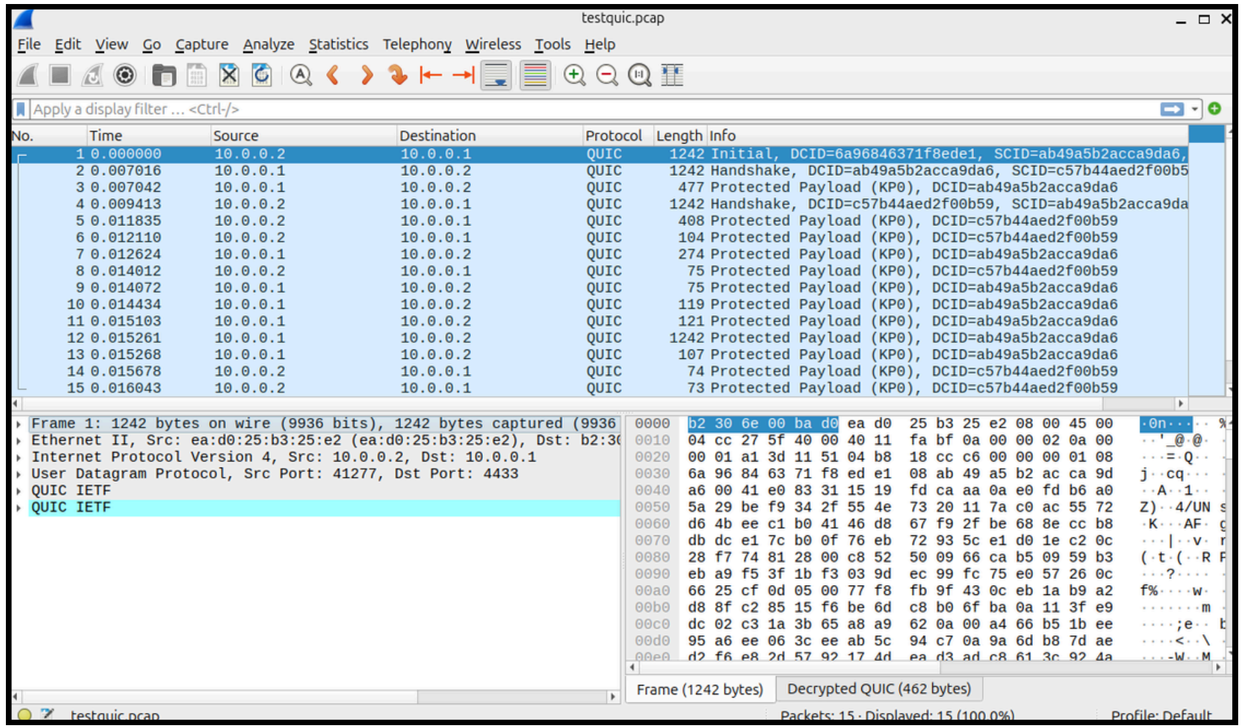
MEDIUM PAYLOAD (B2)



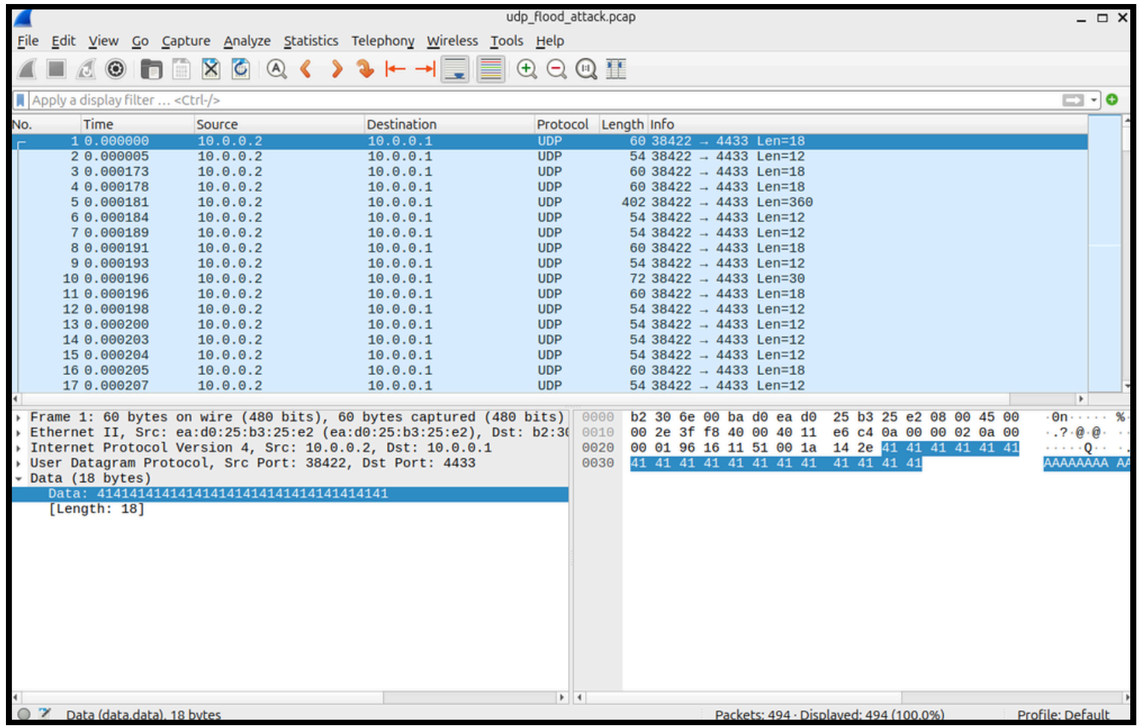
HIGH PAYLOAD (B3)

COMPARISON OF CONDITIONS BEFORE AND AFTER THE ATTACK

CONDITIONS BEFORE THE ATTACK



CONDITIONS AFTER THE ATTACK



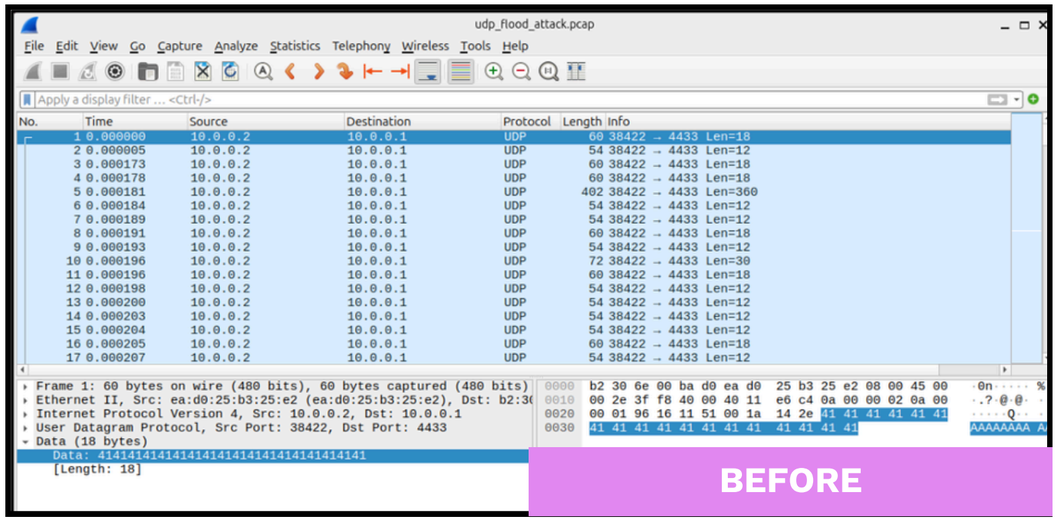
DEFENSE FIREWALL IMPLEMENTATION

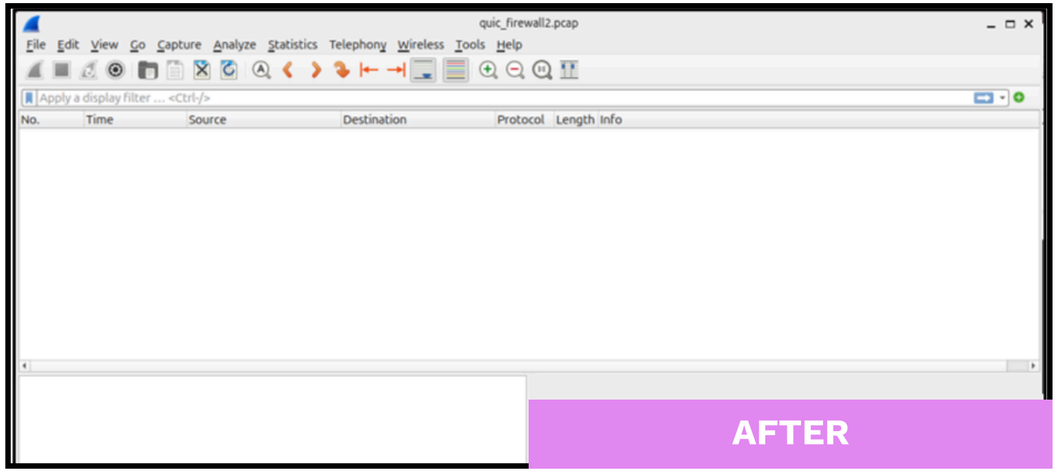
Implementing a Firewall

```
root@ubuntu-capybara:/home/capy/aioquic# iptables -A INPUT -p udp --dport 4433
! --sport 41408 -j DROP
root@ubuntu-capybara:/home/capy/aioquic# iptables -L -n
Chain INPUT (policy ACCEPT)
target     prot opt source               destination
DROP       17   --  0.0.0.0/0            0.0.0.0/0            udp spt:!41408 dpt
:4433
```

- **Current Limitation:** The firewall relies on predefined static rules and requires manual configuration, limiting its ability to adapt to evolving threats. Future development could incorporate adaptive or AI-driven defense mechanisms for automated threat detection and response.

Results





SUMMARY

Analysis Aspect	Scenario	Key Results	Analytical Interpretation
Normal Condition (Baseline)	QUIC without attack.	Bidirectional QUIC traffic (Initial, Handshake, Protected Payload).	QUIC communication operates normally and follows protocol specifications.
Intensity Variation	Low Intensity.	515 packets (10 seconds).	Low packet processing pressure, no significant service disruption.
	Medium Intensity.	1,823 packets (10 seconds).	Increased packet processing load, potential service degradation.
	High Intensity.	314,555 packets (10 seconds).	Volumetric DoS behaviour with extreme packet processing pressure.
Server Behaviour During Attack	UDP Flood	No server response observed.	Attack packets are not valid QUIC packets; traffic is unidirectional.
Payload Variation	Small Payload (±10 bytes).	2,085 packets captured.	Dominant packet processing pressure.
	Medium Payload (±200 bytes).	1,969 packets captured.	Increased bandwidth consumption.
	High Payload (±800 bytes).	1,755 packets captured.	High bandwidth and buffer pressure.
Firewall (Before Mitigation)	UDP Flood active.	QUIC and attack traffic observed.	Server processes both legitimate and malicious traffic.
Firewall (After Mitigation)	Firewall enabled.	No packets captured.	Attack fully blocked.
Security Implications	Overall experiment.	QUIC exposed to UDP-based attacks.	Layered and adaptive mitigation mechanisms are required.


THANK YOU